

notebook_report

July 7, 2026

```
[1]: from data.data_loader import load_data
from data.preprocessing import prepare_datasets, split_data
from features.engineering import FeatureEngineer
from visualization.eda import run_eda, freq_exposition, plot_qualitative_distributions
from models.frequency_model import train_frequency_model
from models.severity_model import train_severity_model, train_best_severity_model
from models.premium_model import build_premium_predictions
from models.frequency_model import evaluate_frequency_model
from models.severity_model import evaluate_severity_model
```

```
c:\Users\Dell\AppData\Local\Programs\Python\Python311\Lib\site-packages\sklearn\utils\_param_validation.py:11: UserWarning: A NumPy version >=1.23.5 and <2.3.0 is required for this version of SciPy (detected version 2.4.2)
```

```
from scipy.sparse import csr_matrix, issparse
```

1 Load data

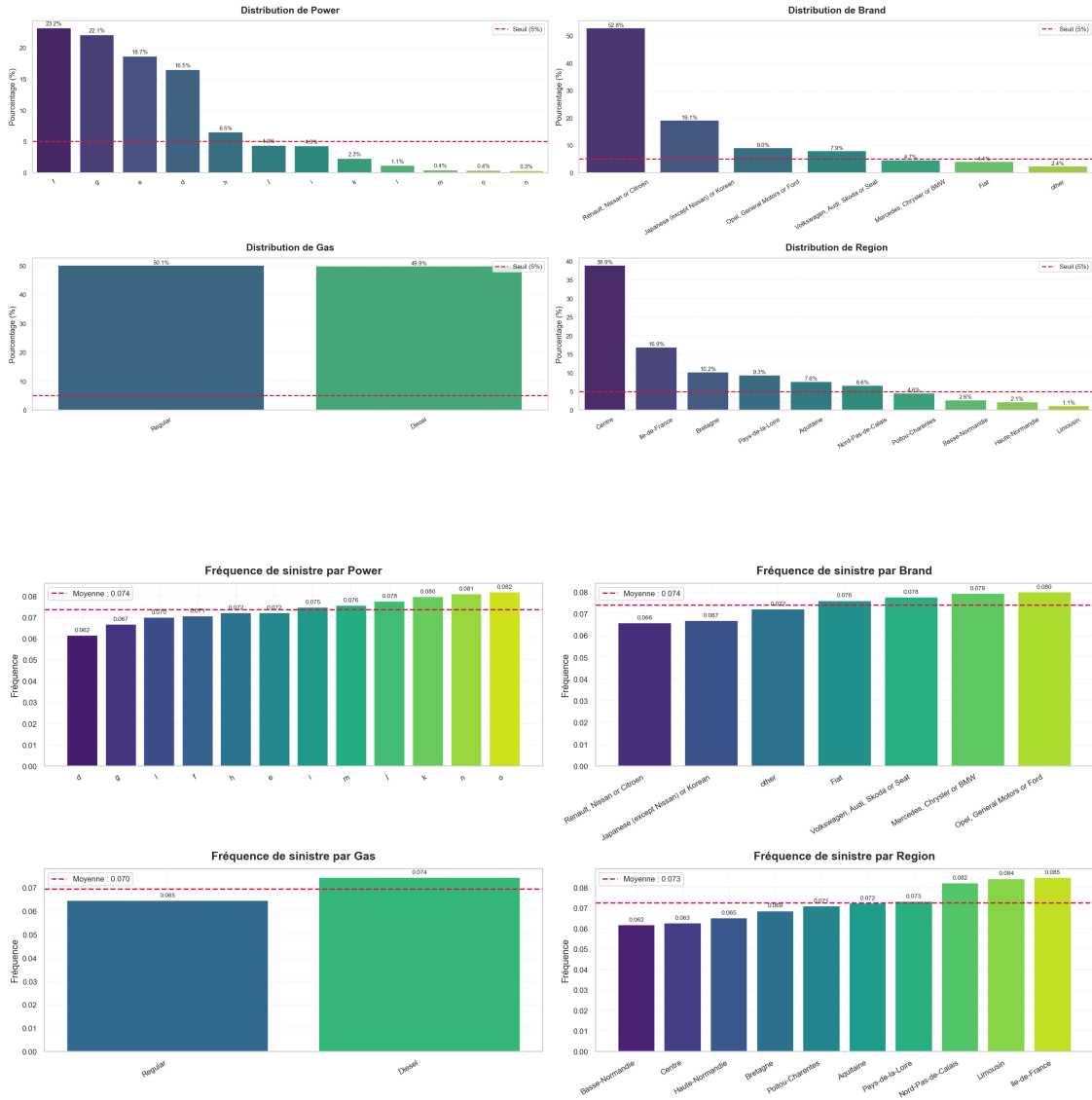
```
[2]: data, data_sev = load_data()

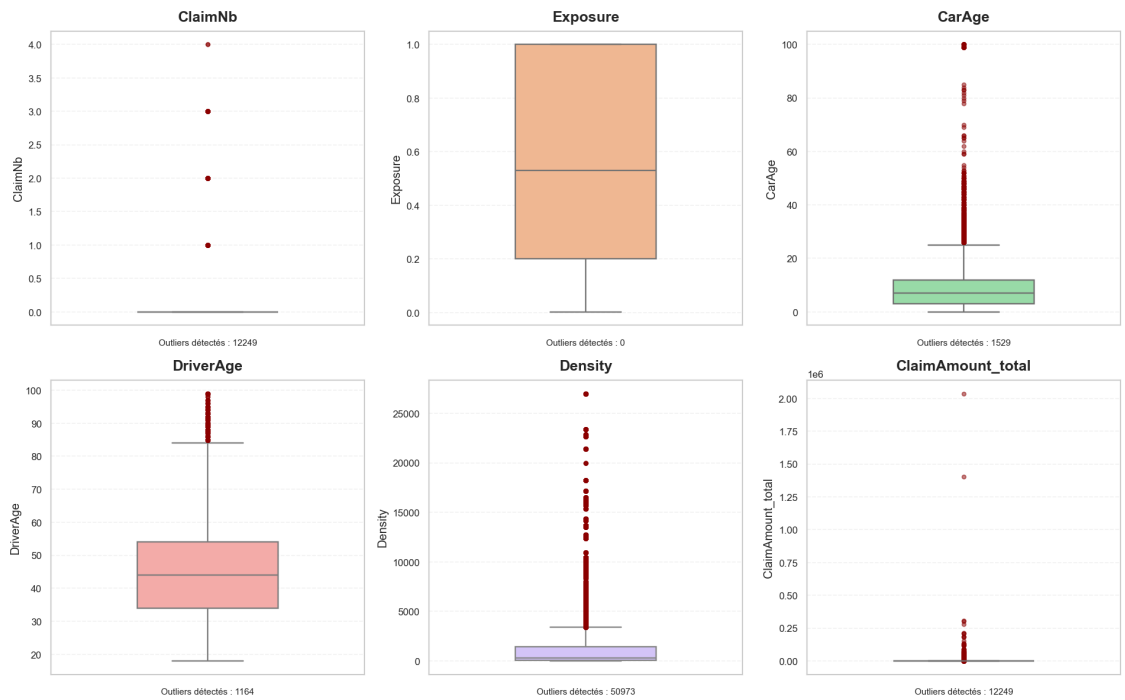
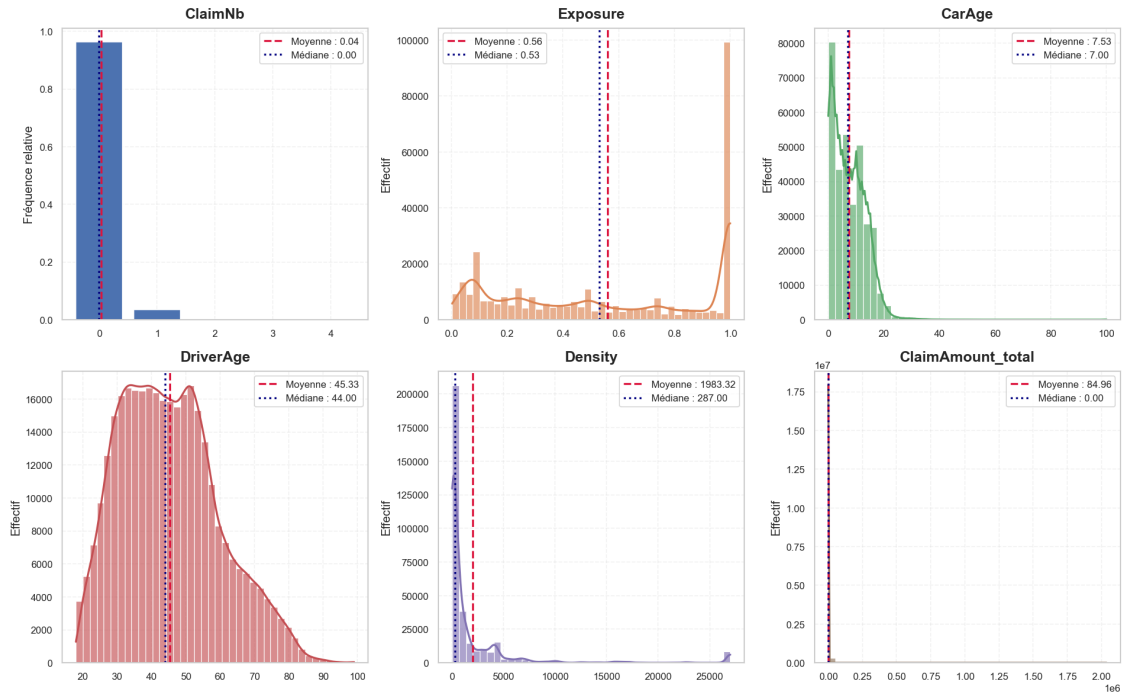
qualitatives_var = data.drop(
    columns=['PolicyID']
).select_dtypes(include=['object', 'category']).columns

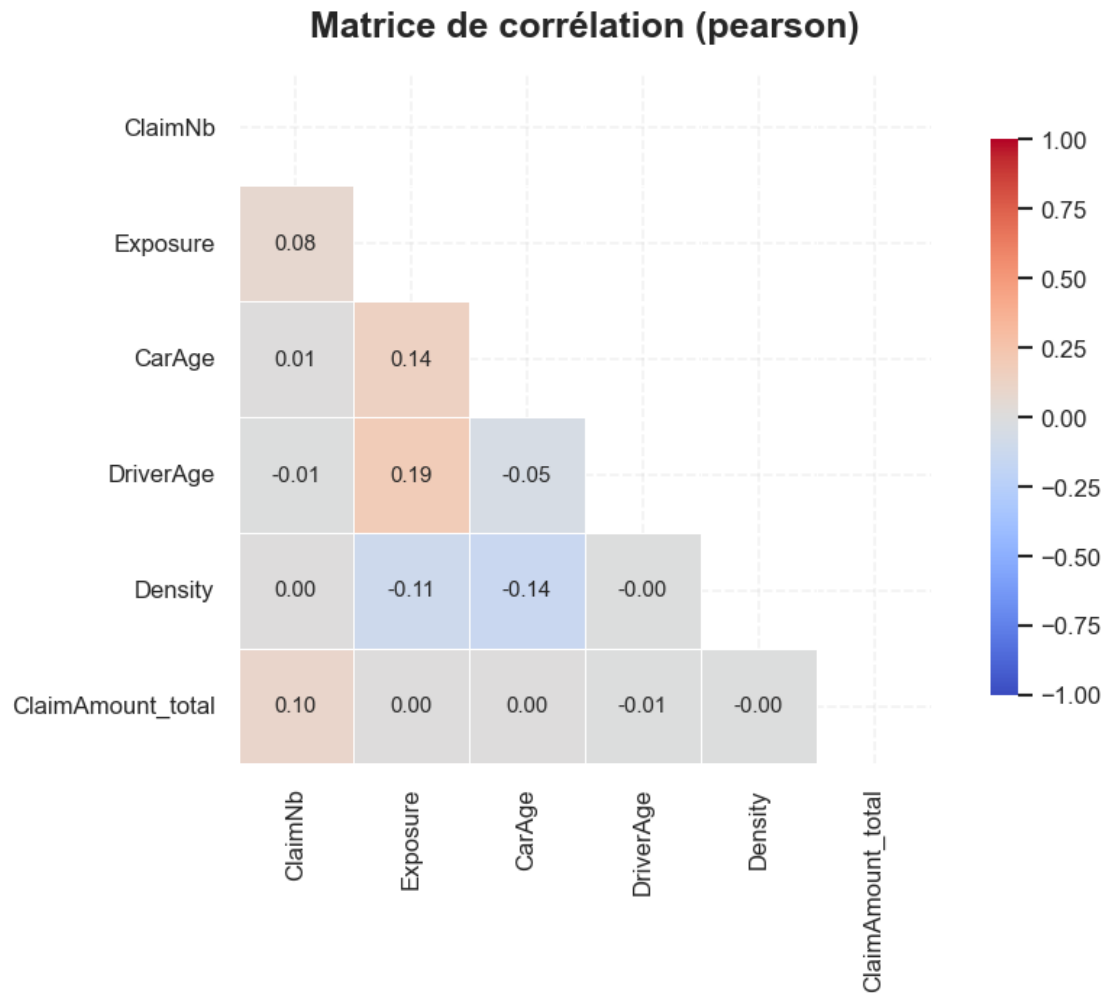
quantitatives_var = data.drop(
    columns=['PolicyID']
).select_dtypes(include=['float64', 'int64']).columns
```

2 EDA

```
[3]: run_eda(
    split_data(data, data_sev)[0],
    qualitatives_var,
    quantitatives_var
)
```







3 Preprocessing

```
[4]: train_freq, test_freq, train_sev, test_sev = (
      prepare_datasets(data, data_sev)
    )
```

4 Feature engineering

```
[5]: fe = FeatureEngineer()

train_freq = fe.fit_transform(train_freq)

test_freq = fe.transform(test_freq)
```

```
train_sev = fe.transform(train_sev)
```

```
test_sev = fe.transform(test_sev)
```

```
[6]: fe.get_discretization_summary()
```

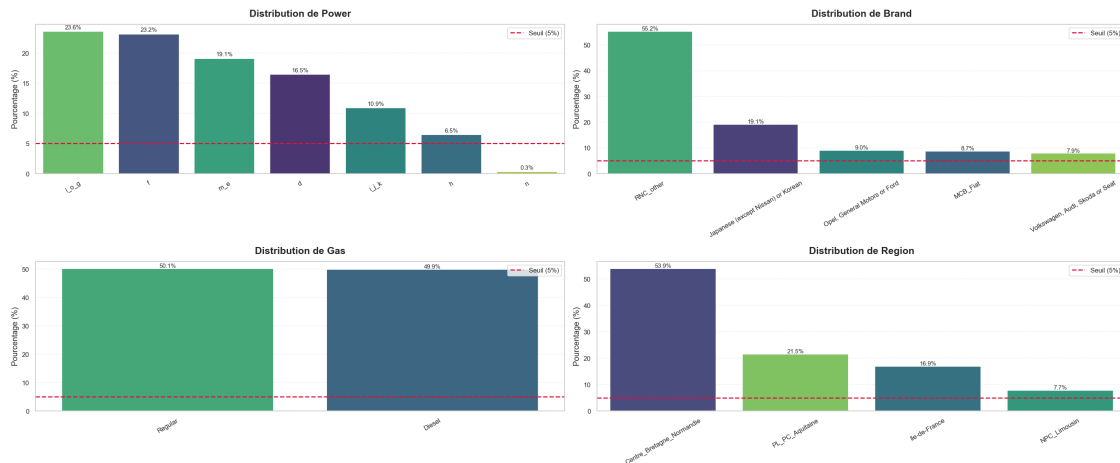
```
[6]:
```

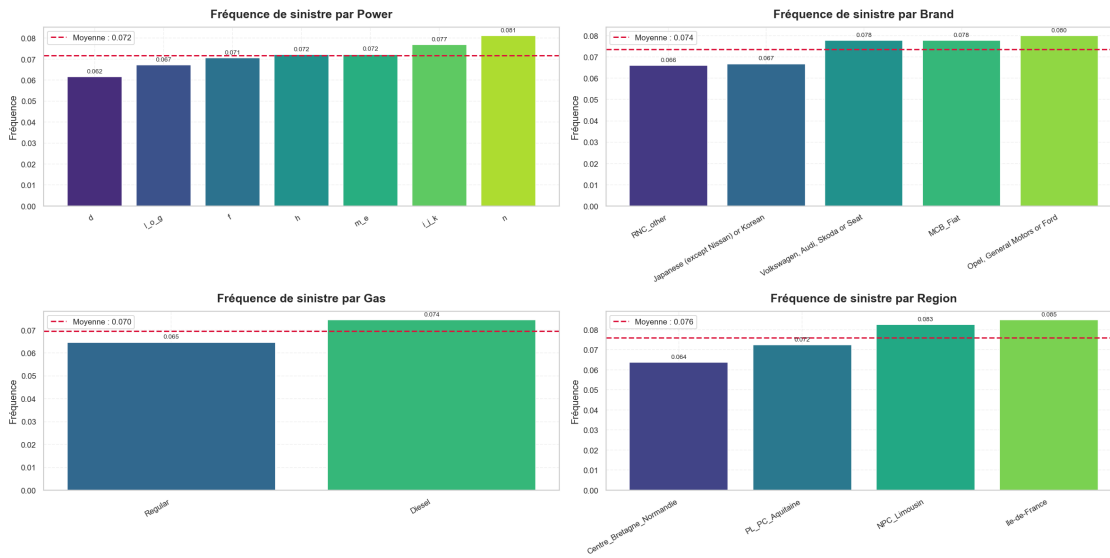
	variable	classe	borne_inf	borne_sup
0	CarAge	CarAge_class_1	0.000000	7.000000
1	CarAge	CarAge_class_2	7.000000	100.000000
2	DriverAge	DriverAge_class_1	18.000000	32.000000
3	DriverAge	DriverAge_class_2	32.000000	40.000000
4	DriverAge	DriverAge_class_3	40.000000	48.000000
5	DriverAge	DriverAge_class_4	48.000000	57.000000
6	DriverAge	DriverAge_class_5	57.000000	99.000000
7	Density	Density_class_1	0.693147	3.931826
8	Density	Density_class_2	3.931826	5.062595
9	Density	Density_class_3	5.062595	6.309918
10	Density	Density_class_4	6.309918	7.784889
11	Density	Density_class_5	7.784889	10.203592

5 Eda post-features engineering

```
[7]: plot_qualitative_distributions(
    train_freq,
    qualitatitives_var,
    save = "_post_fe"
)

freq_exposition(
    train_freq,
    qualitatitives_var,
    save = "_post_fe"
)
```





```
[7]: {'Power':
      sinistres    exposition    freq
      Power
      d          1865  30274.083744  0.061604
      l_o_g      2928  43608.239012  0.067143
      f          3159  44707.183219  0.070660
      h           802  11116.110870  0.072148
      m_e        2636  36481.813389  0.072255
      i_j_k      1431  18602.701240  0.076924
      n           44    542.329240  0.081132,
      'Brand':
      sinistres    exposition    freq
      Brand
      RNC_other          7418  112282.936149  0.066065
      Japanese (except Nissan) or Korean  1665  24914.574899  0.066828
      Volkswagen, Audi, Skoda or Seat    1136  14613.949878  0.077734
      MCB_Fiat           1254  16120.264279  0.077790
      Opel, General Motors or Ford        1392  17400.735509  0.079997,
      'Gas':
      sinistres    exposition    freq
      Gas
      Regular          6141  94970.863860  0.064662
      Diesel           6724  90361.596854  0.074412,
      'Region':
      sinistres    exposition    freq
      Region
      Centre_Bretagne_Normandie          7167  112333.040924  0.063801
      PL_PC_Aquitaine                    2736  37798.145759  0.072385
      NPC_Limousin                        919  11127.876438  0.082585
      Ile-de-France                      2043  24073.397592  0.084865}
```

6 Frequency model

```
[8]: freq_model = train_frequency_model(
    train_freq
)

freq_metrics = evaluate_frequency_model(
    freq_model,
    test_freq
)
```

7 Severity model

```
[9]: sev_model, best_result = train_best_severity_model(
    train_sev
)

sev_metrics = evaluate_severity_model(
    sev_model,
    test_sev
)
```

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```
warnings.warn(  
c:\Users\Dell\AppData\Local\Programs\Python\Python311\Lib\site-  
packages\statsmodels\genmod\generalized_linear_model.py:1923: FutureWarning: The  
bic value is computed using the deviance formula. After 0.13 this will change to  
the log-likelihood based formula. This change has no impact on the relative rank  
of models compared using BIC. You can directly access the log-likelihood version  
using the `bic_llf` attribute. You can suppress this message by calling  
statsmodels.genmod.generalized_linear_model.SET_USE_BIC_LLIF with True to get the  
LLF-based version now or False to retain the deviance version.
```

```
warnings.warn(  

```

8 Pure premium model

```
[10]: premium_data, premium_results = (
        build_premium_predictions(
            test_freq,
            freq_model,
            sev_model
        )
    )

    print(freq_metrics)
    print(sev_metrics)
    print(premium_results)

{'MAE': np.float64(0.0753), 'RMSE': np.float64(0.2065)}
{'MAE': np.float64(1944.1371), 'RMSE': np.float64(8431.6645)}
{'observed_cost': np.float64(6382638.0), 'predicted_premium':
np.float64(6911910.59), 'sp_ratio': np.float64(0.9234), 'premium_metrics':
{'MAE': np.float64(152.93), 'RMSE': np.float64(1693.22)}}

[11]: from pricing.tarif_coefficients import GLMCoefficients
from pricing.pricing_engine import PricingEngine
from pricing.relativities import compute_relativities

coeffs = GLMCoefficients(freq_model, sev_model)
engine = PricingEngine(coeffs)

VARIABLES = {
    "Power": {
        "ref": "d",
        "modalites": ["d", "f", "h", "i_j_k",
            "l_o_g", "m_e", "n"]},
    "Brand": {
        "ref": "Japanese (except Nissan) or Korean",
        "modalites": ["Japanese (except Nissan) or Korean",
            "MCB_Fiat",
            "Opel, General Motors or Ford",
            "RNC_other",
            "Volkswagen, Audi, Skoda or Seat"]},
    "Region": {
        "ref": "Centre_Bretagne_Normandie",
        "modalites": ["Centre_Bretagne_Normandie",
            "Ile-de-France",
            "NPC_Limousin",
            "PL_PC_Aquitaine"]},
    "DriverAge_class": {
        "ref": "DriverAge_class_1",
        "modalites": ["DriverAge_class_1",
            "DriverAge_class_2",
            "DriverAge_class_3",
            "DriverAge_class_4",
            "DriverAge_class_5"]},
```

```

    "CarAge_class": {"ref": "CarAge_class_1",
                     "modalites": ["CarAge_class_1",
                                   "CarAge_class_2"]},
    "Gas": {"ref": "Diesel",
            "modalites": ["Diesel", "Regular"]},
    "Density_class": {"ref": "Density_class_1",
                      "modalites": ["Density_class_1",
                                    "Density_class_2",
                                    "Density_class_3",
                                    "Density_class_4",
                                    "Density_class_5"]},
}

```

```
[12]: rel_table = compute_relativities(coeffs, VARIABLES)
```

```
prime_ref = engine.reference_premium()
```

```

profile = {
    "Power": "d",
    "Brand": "MCB_Fiat",
    "Region": "Centre_Bretagne_Normandie",
    "DriverAge_class": "DriverAge_class_3",
    "CarAge_class": "CarAge_class_1",
    "Gas": "Diesel",
    "Density_class": "Density_class_2"
}

```

```

print(f"Prime de référence : {prime_ref:.2f}")
print(f"Prime pure : {engine.pure_premium(profile)}")
print(f"Tableau de relativities : {rel_table}")

```

Prime de référence : 188.06

Prime pure : {'freq': np.float64(0.06698142713665929), 'sev':
np.float64(1439.146696292323), 'premium': np.float64(96.39609957666816)}

Tableau de relativities : Variable

Modalité	Référence	Rel_Freq \		
0	Power	d	True	1.000000
1	Power	f	False	1.106128
2	Power	h	False	1.132292
3	Power	i_j_k	False	1.242863
4	Power	l_o_g	False	1.083724
5	Power	m_e	False	1.102212
6	Power	n	False	1.265822
7	Brand	Japanese (except Nissan) or Korean	True	1.000000
8	Brand	MCB_Fiat	False	1.298636
9	Brand	Opel, General Motors or Ford	False	1.357797
10	Brand	RNC_other	False	1.192902
11	Brand	Volkswagen, Audi, Skoda or Seat	False	1.252664

12	Region	Centre_Bretagne_Normandie	True	1.000000
13	Region	Ile-de-France	False	1.091248
14	Region	NPC_Limousin	False	1.115861
15	Region	PL_PC_Aquitaine	False	1.060687
16	DriverAge_class	DriverAge_class_1	True	1.000000
17	DriverAge_class	DriverAge_class_2	False	0.675830
18	DriverAge_class	DriverAge_class_3	False	0.772210
19	DriverAge_class	DriverAge_class_4	False	0.735844
20	DriverAge_class	DriverAge_class_5	False	0.642450
21	CarAge_class	CarAge_class_1	True	1.000000
22	CarAge_class	CarAge_class_2	False	0.958122
23	Gas	Diesel	True	1.000000
24	Gas	Regular	False	0.851934
25	Density_class	Density_class_1	True	1.000000
26	Density_class	Density_class_2	False	1.170201
27	Density_class	Density_class_3	False	1.284371
28	Density_class	Density_class_4	False	1.533043
29	Density_class	Density_class_5	False	1.715638

	Rel_Sev	Rel_Combined
0	1.000000	1.000000
1	1.386585	1.533741
2	1.096218	1.241239
3	1.628419	2.023902
4	1.157050	1.253923
5	1.017915	1.121959
6	1.210036	1.531690
7	1.000000	1.000000
8	0.720768	0.936015
9	0.670756	0.910751
10	0.838880	1.000701
11	0.851409	1.066530
12	1.000000	1.000000
13	0.774842	0.845545
14	0.734797	0.819931
15	0.777826	0.825030
16	1.000000	1.000000
17	0.699724	0.472894
18	0.625101	0.482710
19	0.670753	0.493570
20	0.697170	0.447897
21	1.000000	1.000000
22	1.023341	0.980485
23	1.000000	1.000000
24	1.114274	0.949288
25	1.000000	1.000000
26	0.969486	1.134494
27	0.940847	1.208397


```
28 0.821545      1.259465
29 0.814768      1.397847
```

```
[13]: fixed_profile = {
    "Power": "d",
    "Brand": "Japanese (except Nissan) or Korean",
    "Region": "Centre_Bretagne_Normandie",
    "DriverAge_class": "DriverAge_class_3",
    "CarAge_class": "CarAge_class_1",
    "Gas": "Diesel",
    "Density_class": "Density_class_3"
}

powers = VARIABLES["Power"]["modalites"]
brands = VARIABLES["Brand"]["modalites"]

from pricing.tarif_grid import build_grid
grid = build_grid(engine, "Power", "Brand", fixed_profile, powers, brands)

print(grid)
```

	Japanese (except Nissan) or Korean	MCB_Fiat \
d	109.69	102.68
f	168.24	157.48
h	136.16	127.44
i_j_k	222.01	207.81
l_o_g	137.55	128.75
m_e	123.07	115.20
n	168.02	157.27

	Opel, General Motors or Ford	RNC_other \
d	99.90	109.77
f	153.23	168.36
h	124.01	136.25
i_j_k	202.20	222.17
l_o_g	125.27	137.64
m_e	112.09	123.16
n	153.02	168.14

	Volkswagen, Audi, Skoda or Seat
d	116.99
f	179.44
h	145.22
i_j_k	236.78
l_o_g	146.70
m_e	131.26
n	179.20

```
[14]: from models.large_claims_model import run_large_claims_pipeline, \
      ↪compute_tariff_with_large_claim_loading
      from visualization.large_claims_plots import (
          plot_mean_excess_function,
          plot_premium_decomposition
      )
      from models.premium_model import compute_sp_ratio, compute_observed_cost
```

```
[15]: # MODÈLE GROS SINISTRES
      # NB : le pipeline force la cible du GLM attritionnel à
      # severity_capped (sévérité écrêtée au seuil M1) ; seule la
      # partie droite de la formule (les variables) est utilisée.
      large_claims_results = run_large_claims_pipeline(
          train_sev=train_sev,
          test_freq=premium_data,
          formula="""
              severity_capped ~
              C(Power)
              + C(Brand)
              + C(Region)
              + C(DriverAge_class)
              + C(CarAge_class)
              + C(Gas)
              + C(Density_class)
              """,
          claim_amount_col="ClaimAmount_total",
          quantile=0.995
      )

      print("RÉSULTATS LARGE CLAIMS")
      print(
          f"Seuil M1 : "
          f"{large_claims_results['threshold_info']['threshold']:.0f} €"
      )

      print(
          f"P(atypique) : "
          f"{large_claims_results['loading_info']['p_large']:.4%}"
      )

      print(
          f"E[excédent | atypique] : "
          f"{large_claims_results['loading_info']['mean_excess']:.2f} €"
      )
```

```

# PRIME DEUX COMPOSANTES
test_freq_2comp = large_claims_results["test_predictions"].copy()

print("PRIME PURE DEUX COMPOSANTES")
print(
    test_freq_2comp[
        "pure_premium_2comp"
    ].describe()
)

# APPROCHE LOADING SIMPLE
# Chargement atypique unitaire =  $P(\text{atypique}) \times E[\text{excédent} \mid \text{atypique}]$ ,
# appliqué à la prime pure standard (sans écrêtement).
# Attention : cette approche double-compte partiellement les gros
# sinistres (la sévérité standard n'est pas écrêtée) - elle est
# conservée à titre de comparaison avec l'approche deux composantes.
loading_per_claim = (
    large_claims_results["loading_info"]["p_large"]
    * large_claims_results["loading_info"]["mean_excess"]
)

test_freq_loaded = (
    compute_tariff_with_large_claim_loading(
        data=premium_data,
        base_premium_col="pure_premium",
        frequency_col="freq_pred",
        loading_per_claim=loading_per_claim
    )
)

print("PRIME AVEC LOADING")
print(
    test_freq_loaded[
        "pure_premium_loaded"
    ].describe()
)

# COMPARAISON DES APPROCHES
comparison = (
    premium_data[
        "pure_premium"
    ]
    .copy()
)

```

```

comparison["premium_loading"] = (
    test_freq_loaded[
        "pure_premium_loaded"
    ]
)

comparison["premium_2comp"] = (
    test_freq_2comp[
        "pure_premium_2comp"
    ]
)

print("COMPARAISON")
print(
    comparison.mean()
)

#VISUALISATIONS
plot_premium_decomposition(
    data=test_freq_2comp,
    group_col="Density_class",
)

# EXPORT
comparison.to_csv(
    "premium_comparison.csv",
    index=False
)

print("EXPORT TERMINÉ")
print("premium_comparison.csv")

observed_cost_2comp = compute_observed_cost(
    data=test_freq_2comp)
sp_ratio_2comp = compute_sp_ratio(
    observed_cost=observed_cost_2comp,
    predicted_premium=test_freq_2comp["pure_premium_2comp"].sum()
)

observed_cost_loaded = compute_observed_cost(
    data=test_freq_loaded)
sp_ratio_loaded = compute_sp_ratio(
    observed_cost=observed_cost_loaded,
    predicted_premium=test_freq_loaded["pure_premium_loaded"].sum()
)

```

```
)

print(f"SP Ratio 2 composantes : {sp_ratio_2comp:.4f}")
print(f"SP Ratio loading : {sp_ratio_loaded:.4f}")
```

RÉSULTATS LARGE CLAIMS

Seuil M1 : 40,107 €

P(atypique) : 0.4819%

E[excédent | atypique] : 108,716.10 €

PRIME PURE DEUX COMPOSANTES

```
count      82634.000000
mean        85.251461
std         59.798495
min          0.204379
25%         30.812207
50%         83.468467
75%        127.342564
max         440.554118
```

Name: pure_premium_2comp, dtype: float64

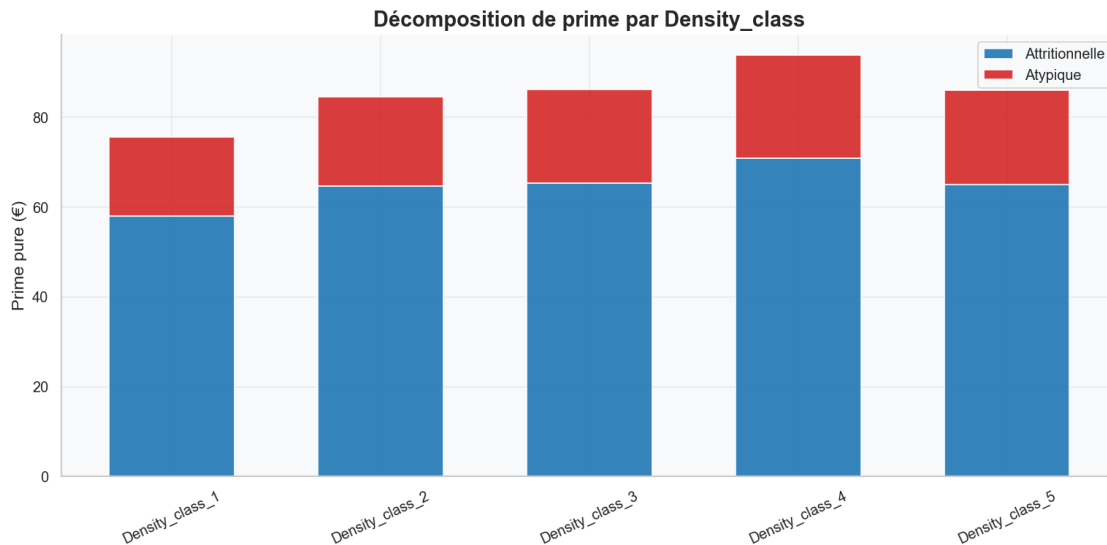
PRIME AVEC LOADING

```
count      82634.000000
mean       104.107091
std        79.191610
min         0.228300
25%         35.899385
50%         97.447939
75%        150.696463
max        593.342369
```

Name: pure_premium_loaded, dtype: float64

COMPARAISON

```
pure_premium      83.644875
premium_loading   104.107091
premium_2comp     85.251461
dtype: float64
```



EXPORT TERMINÉ

premium_comparison.csv

SP Ratio 2 composantes : 0.9060

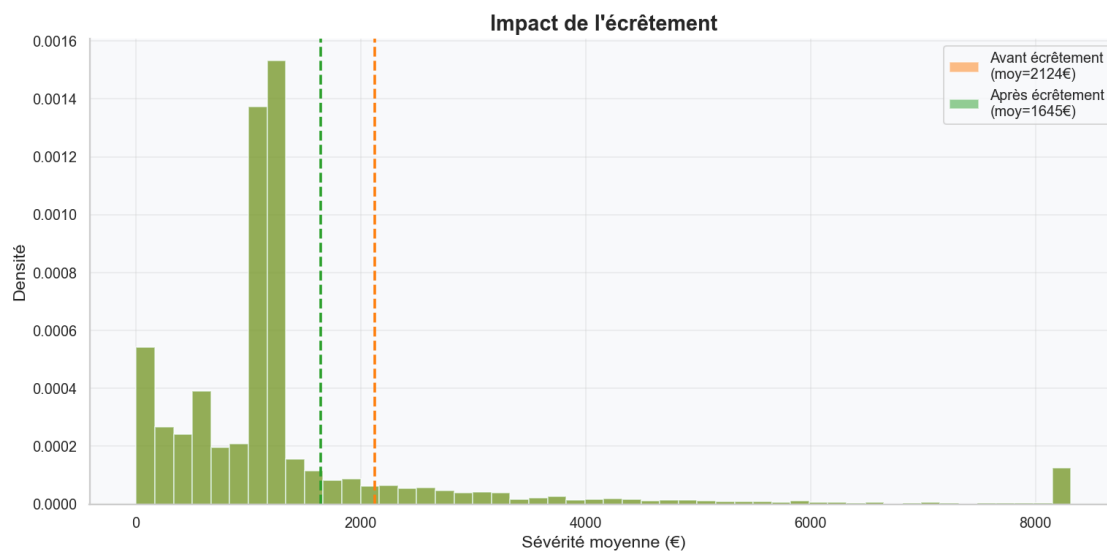
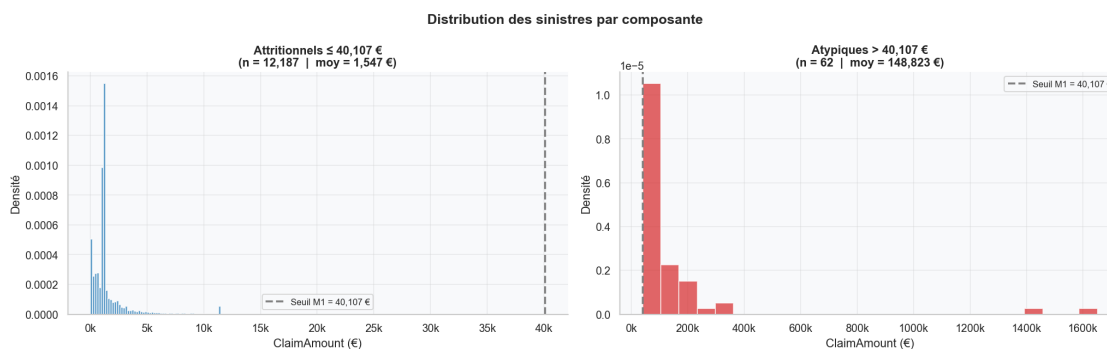
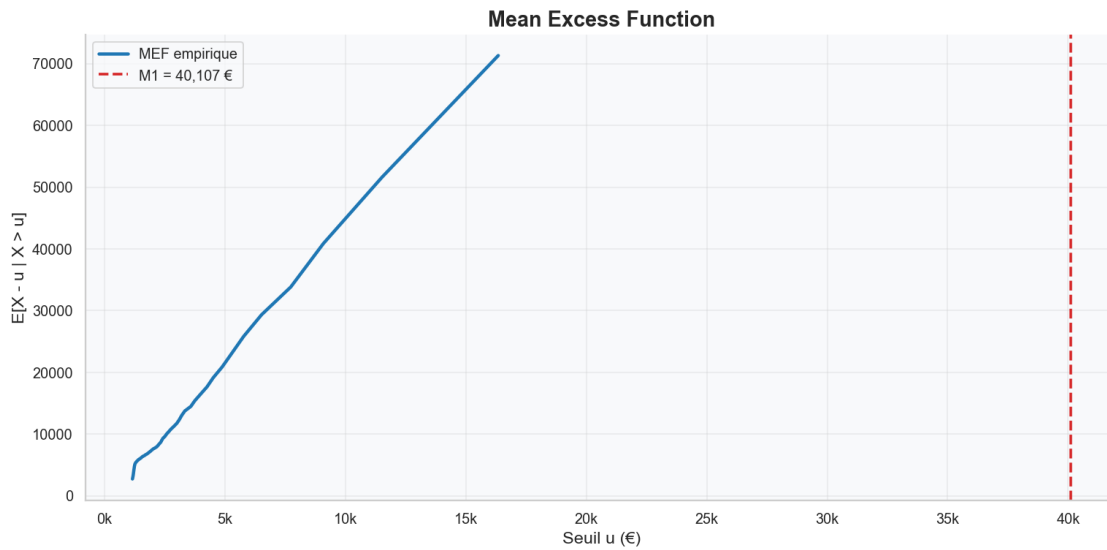
SP Ratio loading : 0.7419

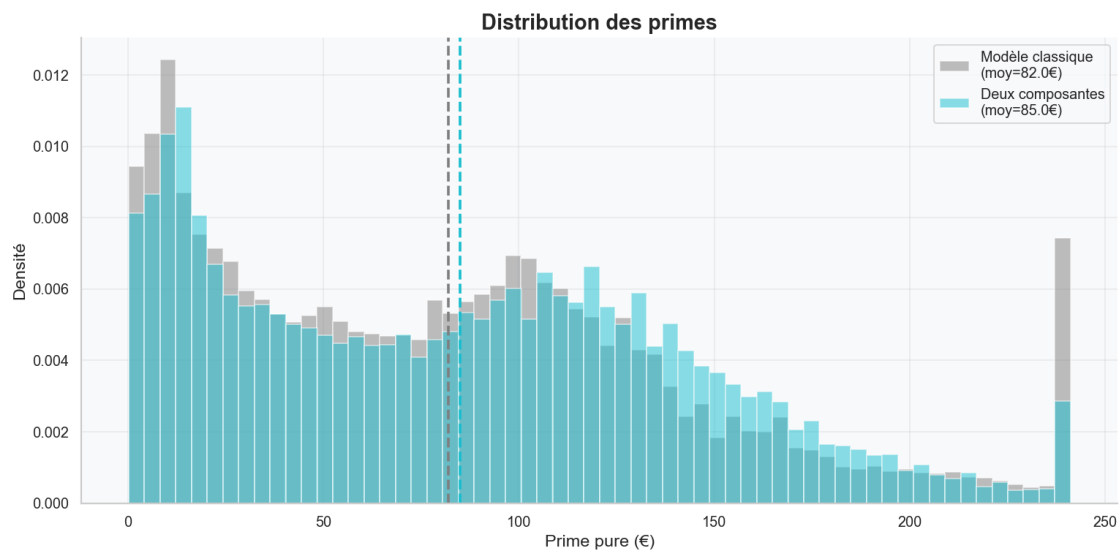
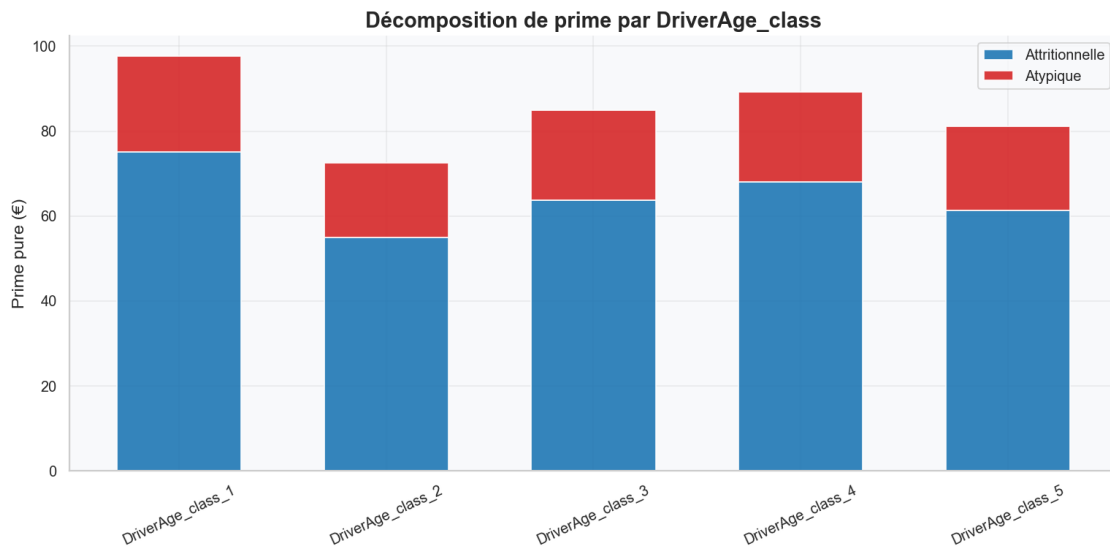
```
[16]: large_claims_results.keys()
```

```
[16]: dict_keys(['threshold_info', 'attritional_claims', 'large_claims',
               'train_capped', 'severity_model', 'loading_info', 'test_predictions'])
```

```
[17]: from visualization.large_claims_plots import run_large_claims_report
```

```
[18]: run_large_claims_report(
        train_sev=large_claims_results["train_capped"],
        test_freq=test_freq_2comp,
        threshold_info=large_claims_results["threshold_info"],
        claim_amount="ClaimAmount_total"
    )
```





9 ML

```
[19]: from data.preprocessing import split_data
      from features.ml_features_engineer import MLFeatureEngineer
```

```
[20]: ml_train_freq, ml_test_freq, ml_train_sev, ml_test_sev = split_data(data, ↵
      ↪data_sev)
```



```

ml_fe = MLFeatureEngineer()
ml_train_freq = ml_fe.fit_transform(ml_train_freq)
ml_test_freq = ml_fe.transform(ml_test_freq)
ml_train_sev = ml_fe.transform(ml_train_sev)
ml_test_sev = ml_fe.transform(ml_test_sev)

```

```

[21]: FEATURES = ml_train_freq.drop(
        columns=["ClaimAmount_total", "Exposure", "ClaimNb", "PolicyID"]
    ).columns.tolist()

```

FEATURES

```

[21]: ['Power',
        'CarAge',
        'DriverAge',
        'Brand',
        'Gas',
        'Region',
        'Density',
        'log_Density']

```

```

[22]: cat_features = ml_train_freq.select_dtypes(
        include=["object", "category"]
    ).columns.tolist()

```

```

for col in cat_features:
    ml_train_freq[col] = (
        ml_train_freq[col]
        .astype("category")
    )

    ml_test_freq[col] = (
        ml_test_freq[col]
        .astype("category")
    )

    ml_train_sev[col] = (
        ml_train_sev[col]
        .astype("category")
    )

    ml_test_sev[col] = (
        ml_test_sev[col]
        .astype("category")
    )
cat_features

```

```
[22]: ['Power', 'Brand', 'Gas', 'Region']
```

```
[23]: from sklearn.model_selection import train_test_split
      from models.ml_frequency_model import LGBFrequencyModel

      # Jeu de validation pour l'early stopping :
      # le nombre d'arbres est choisi par les données,
      # pas fixé a priori.
      ml_tr_freq, ml_val_freq = train_test_split(
          ml_train_freq, test_size=0.2, random_state=42
      )

      # Contrainte de monotonie actuarielle :
      # la fréquence de sinistre croît avec la densité de
      # population (évite les inversions locales indéfendables).
      MONOTONE_CONSTRAINTS = {"Density": +1, "log_Density": +1}

      ml_freq_model = LGBFrequencyModel(
          monotone_constraints=MONOTONE_CONSTRAINTS
      )

      ml_freq_model.fit(
          ml_tr_freq[FEATURES],
          ml_tr_freq["ClaimNb"],
          ml_tr_freq["Exposure"],
          X_val=ml_val_freq[FEATURES],
          y_val=ml_val_freq["ClaimNb"],
          exposure_val=ml_val_freq["Exposure"],
      )

      print(
          f"Arbres retenus (early stopping) : "
          f"{ml_freq_model.model.best_iteration_}"
      )
```

Arbres retenus (early stopping) : 235

```
[24]: from models.large_claims_model import prepare_severity_capping
      from models.ml_severity_model import LGBSeverityModel

      # Écrêtement : même méthodologie que le pipeline GLM.
      # Un GBM ne peut pas apprendre la queue de distribution
      # sur quelques dizaines de sinistres extrêmes : on
      # entraîne sur la sévérité écrêtée et l'excédent est
      # tarifé via un chargement.
      capping_ml = prepare_severity_capping(
          ml_train_sev,
          quantile=0.995,
```

```

        claim_amount_col="ClaimAmount_total"
    )

ml_train_sev_capped = capping_ml["train_capped"]

print(
    f"Seuil M1 : {capping_ml['threshold']:.0f} € | "
    f"P(atypique) : {capping_ml['p_large']:.4%} | "
    f"changement/sinistre : "
    f"{capping_ml['loading_per_claim']:.2f} €"
)

ml_tr_sev, ml_val_sev = train_test_split(
    ml_train_sev_capped, test_size=0.2, random_state=42
)

ml_sev_model = LGBSeverityModel()

ml_sev_model.fit(
    ml_tr_sev[FEATURES],
    ml_tr_sev["severity_capped"],
    claim_nb=ml_tr_sev["ClaimNb"],
    X_val=ml_val_sev[FEATURES],
    y_val=ml_val_sev["severity_capped"],
    claim_nb_val=ml_val_sev["ClaimNb"],
)

print(
    f"Arbres retenus (early stopping) : "
    f"{ml_sev_model.model.best_iteration_}"
)

```

Seuil M1 : 40,107 € | P(atypique) : 0.4819% | changement/sinistre : 523.93 €
 Arbres retenus (early stopping) : 3

```

[25]: from models.ml_evaluate import compute_rebalancing_factor

# Prime pure ML = fréquence attendue × (sévérité écrêtée + changement)
loading_ml = capping_ml["loading_per_claim"]

def ml_pure_premium(df):
    freq_pred = (
        ml_freq_model.predict(df[FEATURES]) * df["Exposure"]
    )
    sev_capped_pred = ml_sev_model.predict(df[FEATURES])
    return freq_pred * (sev_capped_pred + loading_ml)

```

```

# Recalage global : contrairement au GLM Poisson, un GBM
# ne garantit pas l'équilibre agrégé - on force S/P = 1
# sur le train par un facteur multiplicatif unique.
factor_ml = compute_rebalancing_factor(
    ml_train_freq["ClaimAmount_total"].sum(),
    ml_pure_premium(ml_train_freq).sum()
)
print(f"Facteur de recalage ML : {factor_ml:.4f}")

ml_test_freq["pure_premium_ml"] = (
    ml_pure_premium(ml_test_freq) * factor_ml
)

ml_test_freq["freq_pred_ml"] = (
    ml_freq_model.predict(ml_test_freq[FEATURES])
    * ml_test_freq["Exposure"]
)

```

Facteur de recalage ML : 1.0003

10 Modèle Tweedie direct

Alternative à la décomposition fréquence $\times$ sévérité : un seul modèle sur la charge annuelle par police (loi Tweedie = Poisson composé Gamma, variance power $(1, 2)$). Il apprend sur **tout** le portefeuille (413k polices, y compris les non-sinistrées) au lieu de 12k polices sinistrées pour la sévérité.

```

[26]: from models.ml_tweedie_model import LGBTweedieModel

tweedie_model = LGBTweedieModel(
    tweedie_variance_power=1.5,
    monotone_constraints=MONOTONE_CONSTRAINTS
)

tweedie_model.fit(
    ml_tr_freq[FEATURES],
    ml_tr_freq["ClaimAmount_total"],
    ml_tr_freq["Exposure"],
    X_val=ml_val_freq[FEATURES],
    claim_amount_val=ml_val_freq["ClaimAmount_total"],
    exposure_val=ml_val_freq["Exposure"],
)

print(
    f"Arbres retenus (early stopping) : "
    f"{tweedie_model.model.best_iteration_}"
)

```

```

# Recalage S/P = 1 sur le train
premium_train_tw = (
    tweedie_model.predict(ml_train_freq[FEATURES])
    * ml_train_freq["Exposure"]
)

factor_tw = compute_rebalancing_factor(
    ml_train_freq["ClaimAmount_total"].sum(),
    premium_train_tw.sum()
)

print(f"Facteur de recalage Tweedie : {factor_tw:.4f}")

ml_test_freq["pure_premium_tweedie"] = (
    tweedie_model.predict(ml_test_freq[FEATURES])
    * ml_test_freq["Exposure"]
    * factor_tw
)

```

Arbres retenus (early stopping) : 153

Facteur de recalage Tweedie : 1.2930

```

[27]: import pandas as pd
from models.ml_evaluate import model_report, lift_table

# Alignement GLM / ML par PolicyID : mêmes polices de test,
# mais les datasets diffèrent par le feature engineering.
glm_test = test_freq_2comp[
    ["PolicyID", "freq_pred", "pure_premium", "pure_premium_2comp"]
].rename(columns={"freq_pred": "freq_pred_glm"})

eval_df = ml_test_freq.merge(glm_test, on="PolicyID", how="inner")
print(f"Polices de test alignées : {len(eval_df):,}\n")

report = pd.DataFrame([
    model_report(
        "GLM freq x sev",
        eval_df["ClaimNb"], eval_df["freq_pred_glm"],
        eval_df["pure_premium"], eval_df["ClaimAmount_total"]
    ),
    model_report(
        "GLM 2 composantes",
        eval_df["ClaimNb"], eval_df["freq_pred_glm"],
        eval_df["pure_premium_2comp"], eval_df["ClaimAmount_total"]
    ),
    model_report(
        "ML freq x sev (écrêté, recalé)",

```

```

        eval_df["ClaimNb"], eval_df["freq_pred_ml"],
        eval_df["pure_premium_ml"], eval_df["ClaimAmount_total"]
    ),
    model_report(
        "Tweedie direct (recalé)",
        eval_df["ClaimNb"], None,
        eval_df["pure_premium_tweedie"], eval_df["ClaimAmount_total"]
    ),
])

print(report.to_string(index=False))

print("\nLift ML freq x sev (par décile de prime prédite) :")
print(
    lift_table(
        eval_df["ClaimAmount_total"], eval_df["pure_premium_ml"]
    ).to_string(index=False)
)

print("\nLift Tweedie :")
print(
    lift_table(
        eval_df["ClaimAmount_total"], eval_df["pure_premium_tweedie"]
    ).to_string(index=False)
)

```

Polices de test alignées : 82,634

	model	poisson_deviance	gini	sp_ratio
	GLM freq x sev	0.25818	0.1911	0.9234
	GLM 2 composantes	0.25818	0.1803	0.9060
ML freq x sev (écrêté, recalé)		0.25663	0.1830	0.9069
	Tweedie direct (recalé)	NaN	0.1743	0.9126

Lift ML freq x sev (par décile de prime prédite) :

bin	n	pred_mean	obs_mean	ratio_obs_pred
1	8264	5.882813	35.359874	6.010708
2	8263	15.692969	42.989713	2.739425
3	8263	30.271390	41.351204	1.366016
4	8264	49.019226	98.405252	2.007483
5	8263	70.319402	77.304006	1.099327
6	8263	91.233734	63.843035	0.699774
7	8264	109.396119	84.815344	0.775305
8	8263	126.088028	130.266852	1.033142
9	8263	147.050289	77.557425	0.527421
10	8264	206.779220	120.502178	0.582758

Lift Tweedie :

bin	n	pred_mean	obs_mean	ratio_obs_pred
1	8264	5.303498	31.673766	5.972240
2	8263	14.380544	47.830328	3.326044
3	8263	27.774569	49.237565	1.772757
4	8264	44.849610	84.329501	1.880273
5	8263	63.894790	75.345153	1.179207
6	8263	82.779369	83.135060	1.004297
7	8264	101.088491	113.159850	1.119414
8	8263	120.682812	80.180322	0.664389
9	8263	146.067924	93.651095	0.641148
10	8264	239.503617	113.851767	0.475366

```
[28]: from models.ml_evaluate import lorenz_curve, gini
from visualization.ml_plots import (
    plot_lift,
    plot_lorenz_curves,
    plot_feature_importance,
    plot_pdp,
    plot_shap_summary,
)

# Lorenz : pouvoir de segmentation des quatre tarifs
curves = {}
for name, col in [
    ("GLM freq x sev", "pure_premium"),
    ("GLM 2 composantes", "pure_premium_2comp"),
    ("ML freq x sev", "pure_premium_ml"),
    ("Tweedie", "pure_premium_tweedie"),
]:
    x, y = lorenz_curve(eval_df["ClaimAmount_total"], eval_df[col])
    g = gini(eval_df["ClaimAmount_total"], eval_df[col])
    curves[name] = (x, y, g)

plot_lorenz_curves(curves)

# Lift / calibration par décile
plot_lift(
    lift_table(
        eval_df["ClaimAmount_total"], eval_df["pure_premium_ml"]
    ),
    "ml_freq_sev"
)
plot_lift(
    lift_table(
        eval_df["ClaimAmount_total"], eval_df["pure_premium_tweedie"]
    ),
    "tweedie"
```

```

)

# Importance des variables (gain)
plot_feature_importance(ml_freq_model.model, "frequence_lgbm")
plot_feature_importance(tweedie_model.model, "tweedie_lgbm")

# Partial Dependence : effets marginaux moyens,
# à confronter aux relativités du GLM
plot_pdp(
    ml_freq_model.predict,
    ml_test_freq[FEATURES],
    ["DriverAge", "Density", "CarAge", "Brand"],
    "frequence_lgbm",
    ylabel="Fréquence annuelle prédite"
)

# SHAP (TreeSHAP natif LightGBM, échelle log)
plot_shap_summary(
    ml_freq_model.model, ml_test_freq[FEATURES], "frequence_lgbm"
)
plot_shap_summary(
    tweedie_model.model, ml_test_freq[FEATURES], "tweedie_lgbm"
)

```